

Linear Statistical Models - w5 GA

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1. Solution to Question 1

(i). Find the least square estimates $\hat{\beta}_0$ and $\hat{\beta}_1$

We want to find $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of squared residuals, S :

$$S(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

We take the partial derivatives with respect to $\hat{\beta}_0$ and $\hat{\beta}_1$ and set them to zero.

First-Order Condition w.r.t. $\hat{\beta}_0$:

$$\begin{aligned} \frac{\partial S}{\partial \hat{\beta}_0} &= \sum_{i=1}^n 2(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(-1) = 0 \\ \implies -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \\ \implies \sum_{i=1}^n y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n x_i &= 0 \\ \implies \frac{1}{n} \sum_{i=1}^n y_i - \hat{\beta}_0 - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n x_i &= 0 \\ \implies \bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} &= 0 \\ \implies \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} &\quad (\text{Equation 1}) \end{aligned}$$

First-Order Condition w.r.t. $\hat{\beta}_1$:

$$\begin{aligned} \frac{\partial S}{\partial \hat{\beta}_1} &= \sum_{i=1}^n 2(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(-x_i) = 0 \\ \implies -2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \\ \implies \sum_{i=1}^n (x_i y_i - \hat{\beta}_0 x_i - \hat{\beta}_1 x_i^2) &= 0 \\ \implies \sum_{i=1}^n x_i y_i - \hat{\beta}_0 \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= 0 \quad (\text{Equation 2}) \end{aligned}$$

Solve for $\hat{\beta}_1$: Substitute (Equation 1) into (Equation 2):

$$\begin{aligned}
\sum x_i y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) \sum x_i - \hat{\beta}_1 \sum x_i^2 &= 0 \\
\sum x_i y_i - \bar{y} \sum x_i + \hat{\beta}_1 \bar{x} \sum x_i - \hat{\beta}_1 \sum x_i^2 &= 0 \\
\sum x_i y_i - \bar{y}(n\bar{x}) + \hat{\beta}_1(n\bar{x}^2) - \hat{\beta}_1 \sum x_i^2 &= 0 \\
\hat{\beta}_1(n\bar{x}^2 - \sum x_i^2) &= n\bar{x}\bar{y} - \sum x_i y_i \\
\hat{\beta}_1(\sum x_i^2 - n\bar{x}^2) &= \sum x_i y_i - n\bar{x}\bar{y} \\
\hat{\beta}_1 &= \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}
\end{aligned}$$

Using the standard identities $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}$ and $\sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$, we get the final forms:

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}} \quad \text{and} \quad \boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

(ii). Prove that $\hat{\beta}_1$ and $\hat{\beta}_0$ are unbiased

Let $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$. We can rewrite $\hat{\beta}_1$ as:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum (x_i - \bar{x})y_i - \bar{y} \sum (x_i - \bar{x})}{S_{xx}}$$

Since $\sum (x_i - \bar{x}) = 0$, this simplifies to:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})y_i}{S_{xx}}$$

Substitute the true model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$:

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum (x_i - \bar{x})(\beta_0 + \beta_1 x_i + \varepsilon_i)}{S_{xx}} \\
&= \frac{\sum (x_i - \bar{x})\beta_0 + \sum (x_i - \bar{x})\beta_1 x_i + \sum (x_i - \bar{x})\varepsilon_i}{S_{xx}} \\
&= \frac{\beta_0 \sum (x_i - \bar{x}) + \beta_1 \sum (x_i - \bar{x})x_i + \sum (x_i - \bar{x})\varepsilon_i}{S_{xx}}
\end{aligned}$$

Using $\sum (x_i - \bar{x}) = 0$ and $\sum (x_i - \bar{x})x_i = \sum (x_i - \bar{x})(x_i - \bar{x}) = S_{xx}$:

$$\hat{\beta}_1 = \frac{0 + \beta_1 S_{xx} + \sum (x_i - \bar{x})\varepsilon_i}{S_{xx}} = \beta_1 + \frac{\sum (x_i - \bar{x})\varepsilon_i}{S_{xx}}$$

(a) $\hat{\beta}_1$ is unbiased: Taking the expectation (treating x_i as fixed):

$$\begin{aligned}
E[\hat{\beta}_1] &= E\left[\beta_1 + \frac{\sum (x_i - \bar{x})\varepsilon_i}{S_{xx}}\right] \\
&= \beta_1 + \frac{\sum (x_i - \bar{x})E[\varepsilon_i]}{S_{xx}}
\end{aligned}$$

Given $E[\varepsilon_i] = 0$:

$$E[\hat{\beta}_1] = \beta_1 + \frac{\sum (x_i - \bar{x})(0)}{S_{xx}} = \beta_1 \quad \blacksquare$$

(b) $\hat{\beta}_0$ is unbiased:

$$\begin{aligned}
E[\hat{\beta}_0] &= E[\bar{y} - \hat{\beta}_1 \bar{x}] \\
&= E[\bar{y}] - E[\hat{\beta}_1] \bar{x}
\end{aligned}$$

First, $E[\bar{y}] = E[\frac{1}{n} \sum y_i] = \frac{1}{n} \sum E[y_i] = \frac{1}{n} \sum (\beta_0 + \beta_1 x_i) = \frac{1}{n} (n\beta_0 + \beta_1 \sum x_i) = \beta_0 + \beta_1 \bar{x}$. Substitute this and $E[\hat{\beta}_1] = \beta_1$ back:

$$E[\hat{\beta}_0] = (\beta_0 + \beta_1 \bar{x}) - (\beta_1) \bar{x} = \beta_0 \quad \blacksquare$$

(iii). Prove the variances

We use $\text{Var}(\varepsilon_i) = \sigma^2$ and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.

(a) $\text{Var}(\hat{\beta}_1)$: Let $k_i = \frac{(x_i - \bar{x})}{S_{xx}}$. From (ii), we have $\hat{\beta}_1 = \beta_1 + \sum k_i \varepsilon_i$.

$$\begin{aligned} \text{Var}(\hat{\beta}_1) &= \text{Var}(\beta_1 + \sum k_i \varepsilon_i) = \text{Var}(\sum k_i \varepsilon_i) \\ &= \sum \text{Var}(k_i \varepsilon_i) \quad (\text{due to independence}) \\ &= \sum k_i^2 \text{Var}(\varepsilon_i) = \sum k_i^2 \sigma^2 \\ &= \sigma^2 \sum \left(\frac{x_i - \bar{x}}{S_{xx}} \right)^2 = \sigma^2 \frac{\sum (x_i - \bar{x})^2}{S_{xx}^2} \\ &= \sigma^2 \frac{S_{xx}}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}} \end{aligned}$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

(b) $\text{Var}(\hat{\beta}_0)$:

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) \\ &= \text{Var}(\bar{y}) + \text{Var}(\hat{\beta}_1 \bar{x}) - 2\text{Cov}(\bar{y}, \hat{\beta}_1 \bar{x}) \\ &= \text{Var}(\bar{y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\bar{x} \text{Cov}(\bar{y}, \hat{\beta}_1) \end{aligned}$$

We find each term:

$$1. \text{Var}(\bar{y}) = \text{Var}(\frac{1}{n} \sum y_i) = \frac{1}{n^2} \sum \text{Var}(y_i) = \frac{1}{n^2} \sum \text{Var}(\varepsilon_i) = \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}.$$

$$2. \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} \quad (\text{from part a}).$$

$$3. \text{Cov}(\bar{y}, \hat{\beta}_1) = \text{Cov}(\bar{\varepsilon}, \sum k_j \varepsilon_j) = \text{Cov}(\sum \frac{1}{n} \varepsilon_i, \sum k_j \varepsilon_j)$$

(Since $\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{\varepsilon}$ and constants drop from covariance)

$$= \sum_{i=1}^n \frac{1}{n} k_i \text{Var}(\varepsilon_i) = \sum \frac{1}{n} k_i \sigma^2 = \frac{\sigma^2}{n} \sum k_i$$

$$\text{Since } \sum k_i = \sum \frac{(x_i - \bar{x})}{S_{xx}} = \frac{1}{S_{xx}} \sum (x_i - \bar{x}) = 0, \text{ we have } \text{Cov}(\bar{y}, \hat{\beta}_1) = 0.$$

Substituting back:

$$\text{Var}(\hat{\beta}_0) = \frac{\sigma^2}{n} + \bar{x}^2 \left(\frac{\sigma^2}{S_{xx}} \right) - 2\bar{x}(0)$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)$$

2. Solution to Question 2

First, we calculate summary statistics for the given data:

x	y
2	6
4	8
1	2

- $n = 3$
- $\sum x_i = 2 + 4 + 1 = 7 \implies \bar{x} = 7/3$
- $\sum y_i = 6 + 8 + 2 = 16 \implies \bar{y} = 16/3$
- $\sum x_i^2 = 2^2 + 4^2 + 1^2 = 4 + 16 + 1 = 21$
- $\sum y_i^2 = 6^2 + 8^2 + 2^2 = 36 + 64 + 4 = 104$
- $\sum x_i y_i = (2)(6) + (4)(8) + (1)(2) = 12 + 32 + 2 = 46$

Now, we calculate the corrected sums of squares:

$$\begin{aligned}
 S_{xx} &= \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 21 - 3\left(\frac{7}{3}\right)^2 = 21 - \frac{49}{3} = \frac{63 - 49}{3} = \frac{14}{3} \\
 S_{yy} &= \sum (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2 = 104 - 3\left(\frac{16}{3}\right)^2 = 104 - \frac{256}{3} = \frac{312 - 256}{3} = \frac{56}{3} \\
 S_{xy} &= \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y} = 46 - 3\left(\frac{7}{3}\right)\left(\frac{16}{3}\right) = 46 - \frac{112}{3} = \frac{138 - 112}{3} = \frac{26}{3}
 \end{aligned}$$

(i). Find the least square estimates for $\beta_{\sim}$

The vector $\beta_{\sim}$ contains β_0 and β_1 .

$$\begin{aligned}
 \hat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{26/3}{14/3} = \frac{26}{14} = \frac{13}{7} \\
 \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} = \frac{16}{3} - \left(\frac{13}{7}\right)\left(\frac{7}{3}\right) = \frac{16}{3} - \frac{13}{3} = \frac{3}{3} = 1
 \end{aligned}$$

So the vector of estimates is $\hat{\beta}_{\sim} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 13/7 \end{pmatrix}$.

(ii). Calculate the value of RSS (Residual Sum of Squares)

RSS can be calculated as $RSS = S_{yy} - \hat{\beta}_1 S_{xy}$.

$$\begin{aligned}
 RSS &= \frac{56}{3} - \left(\frac{13}{7}\right)\left(\frac{26}{3}\right) \\
 &= \frac{56}{3} - \frac{338}{21} \\
 &= \frac{56 \cdot 7}{21} - \frac{338}{21} = \frac{392 - 338}{21} = \frac{54}{21}
 \end{aligned}$$

$$\boxed{RSS = \frac{18}{7} \approx 2.571}$$

(iii). Calculate the value of TSS (Total Sum of Squares)

The Total Sum of Squares (TSS) is equal to S_{yy} .

$$TSS = S_{yy} = \frac{56}{3} \approx 18.667$$

(iv). Find the value of R^2 and interpret the result

R^2 is the coefficient of determination.

$$\begin{aligned} R^2 &= 1 - \frac{RSS}{TSS} \\ &= 1 - \frac{18/7}{56/3} = 1 - \left(\frac{18}{7} \cdot \frac{3}{56} \right) \\ &= 1 - \frac{54}{392} = 1 - \frac{27}{196} \\ &= \frac{196 - 27}{196} = \frac{169}{196} \end{aligned}$$

$$R^2 = \frac{169}{196} \approx 0.8622$$

Interpretation: The value $R^2 \approx 0.8622$ means that approximately **86.22%** of the total variation in the y values is explained by the linear relationship with the x values. This indicates a very strong linear fit.